

Project Name: Autonomous AI-Powered SOC Analytics System

Course: AI in Cybersecurity based on NVIDIA Morpheus

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Video demonstration: <https://youtu.be/20HiVw677Bo>

1. Executive Summary

This project presents an autonomous Security Operations Center (SOC) analytics prototype designed to alleviate the cognitive burden on human analysts. By integrating a multi-agent framework with automated anomaly detection, we demonstrate a robust pipeline capable of ingesting network logs, identifying high-risk events, and generating professional incident response reports mapped to the MITRE ATT&CK framework.

2. Problem Statement:

Modern Security Operations Centers (SOCs) are currently overwhelmed by the exponential increase in network data and security log volumes. Human analysts are faced with the impossible task of manually triaging thousands of alerts daily, leading to alert fatigue, increased response times, and the potential for critical threats to go unnoticed. Traditional SIEM systems, while capable of logging, often lack the contextual intelligence required to distinguish between routine noise and sophisticated, low-and-slow attack patterns.

To address this critical gap, we have developed an autonomous SOC analytics prototype that integrates machine learning-based detection with generative AI agents. Our solution automates the "Detection-to-Analysis" pipeline by first identifying high-risk statistical anomalies in network traffic and then delegating the interpretation of these events to specialized AI agents. By utilizing a multi-agent architecture that maps findings directly to the MITRE ATT&CK framework, our system significantly reduces the

cognitive load on human analysts, providing immediate, actionable threat intelligence and structured mitigation recommendations.

3. System Architecture & Chosen Technologies:

Our architecture is designed for modularity and scalability, leveraging containerized orchestration:

- **Agent Orchestration:** We employed the Microsoft Agent Framework to manage the interaction between the "Tier 1 Analyst" (triage) and the "CTI Expert" (deep analysis).
- **Data Pipeline:** The system utilizes a Kafka-based pipeline for real-time log ingestion, ensuring modular decoupling of data processing from the analysis logic.
- **Containerization:** The entire solution is Dockerized, ensuring environment consistency and rapid deployment.
- **LLM Integration:** The agents communicate via the Groq API (using Llama-3.3-70b), providing high-speed inference for real-time threat analysis.

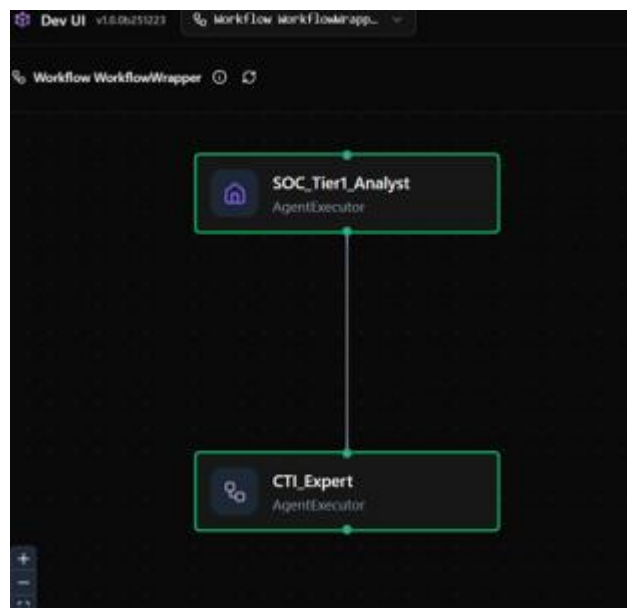


Figure 1: Multi-agent architecture demonstrating the logical flow between the SOC Tier 1 Analyst and the CTI Expert agent

4. Methodology (The MITRE ATT&CK Framework):

In accordance with the project requirements, we adopted the MITRE ATT&CK framework as our universal threat language. Our detection tool flags statistical anomalies based on a risk score threshold. Once triggered, the CTI Expert agent is programmed to automatically interpret these

findings and assign the appropriate **Tactic** and **Technique ID**, ensuring that the output is not just raw data, but actionable intelligence ready for an Incident Response team.

5. Results & Demonstration:

Analysis of Figure 1: The demonstration shows an automated execution flow where the Tier 1 Analyst detected an RDP Brute Force attempt (Risk Score: 95). The CTI Expert agent successfully mapped this event to MITRE ATT&CK Technique T1110 (Brute Force) and Tactic Credential Access. The generated Incident Response Report includes immediate mitigation recommendations, confirming the success of the autonomous pipeline.

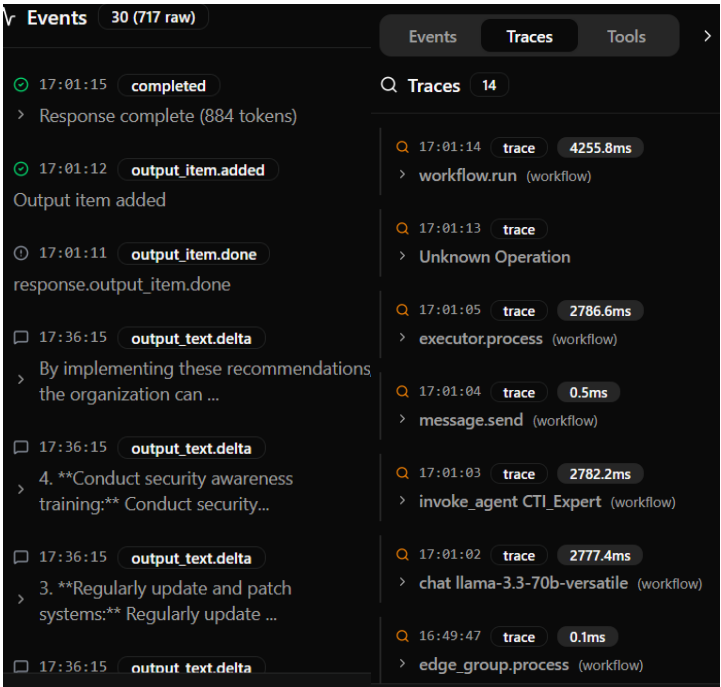


Figure 2 : Execution timeline confirming successful automated process completion.

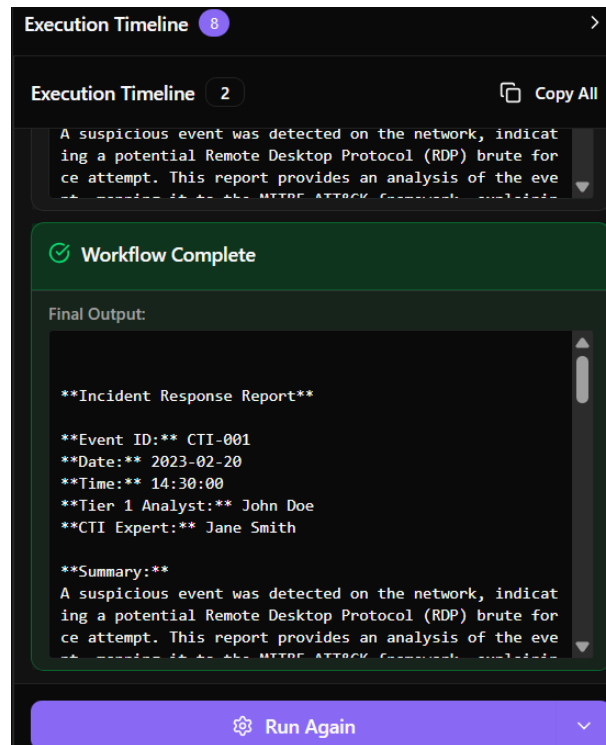


Figure 3: Incident Response Report generated by the CTI Expert agent, including MITRE ATT&CK mapping and mitigation steps.

6. Discussion & Conclusions:

This prototype successfully proves that it is possible to integrate multiple components of the AI-in-cybersecurity ecosystem—simulated log processing, multi-agent AI, and threat intelligence standards—into a unified workflow. The project demonstrates the viability of automating routine SOC tasks. Future work could include integrating live SIEM feeds (e.g., Splunk) to move from a simulation to a production-grade deployment.